FILTER DESIGN TEST CASE

Lattice Filter - WiFi 5 GHz for Millimeter-Wave Beamforming
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0011
Revision:	Rev C.7
Author:	Dr. Pierre Dubois
Reviewer:	Dr. Olaf Bergstrom
Design Center:	Cedar Rapids Lab
Created Date:	2024-10-06
Last Modified:	2024-12-02
Status:	CONDITIONAL PASS
Priority:	P1 - Critical

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Lattice Filter filter targeting WiFi 5 GHz for Millimeter-Wave Beamforming. The filter is designed on AlN on Si substrate with a target center frequency of 4585.8 MHz and bandwidth of 80.4 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Millimeter-Wave Beamforming module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Lattice Filter
- Frequency Band: WiFi 5 GHz
- Application: Millimeter-Wave Beamforming
- Substrate: AlN on Si
- Package: QFN (Quad Flat No-Lead)
- Design Tool: Synopsys CustomSim
- Design Center: Cedar Rapids Lab

This specification applies to all variants of the Lattice Filter design targeting WiFi 5 GHz. Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 4585.8 MHz
Bandwidth (3 dB): 80.4 MHz
Insertion Loss Target: <= 2.3 dB
Return Loss Target: >= 13.0 dB
Out-of-Band Rejection: >= 29.0 dB
Isolation (Duplexer): >= 43.0 dB
Q Factor: 13422
Temperature Coefficient:-21.0 ppm/°C
Die Size: 1.8 x 1.7 mm
Wafer Size: 6-inch
Number of Resonators: 4
Electrode Material: Mo

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Synopsys CustomSim and load the Lattice Filter template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 4585.8 MHz, BW 80.4 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (QFN (Quad Flat No-Lead)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on Si).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 13757 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.3 dB
- Return loss in passband shall be >= 13.0 dB
- Out-of-band rejection at +/- 121 MHz offset >= 29.0 dB
- Group delay variation across passband shall be <= 5 ns
- Temperature drift over -40°C to +85°C shall be <= 12.0 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 95%

6. TEST RESULTS

Measurement Date: 2024-12-02
Devices Tested: 48
Insertion Loss (meas): 2.02 dB
Return Loss (meas): 15.0 dB
Rejection (meas): 32.4 dB
Isolation (meas): 48.0 dB
Q Factor (meas): 13422
Group Delay Variation: 6.0 ns
Power Handling: +28 dBm
Die Yield: 94.6%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

a) The Lattice Filter design demonstrated acceptable performance for WiFi 5 GHz.
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Group delay flatness improved compared to previous revision.
e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

Author:	Dr. Pierre Dubois	Date: 2024-12-02
Reviewer:	Dr. Olaf Bergstrom	Date: 2024-12-07
Approver:	Dr. Yuko Ishida	Date: 2024-12-10

END OF DOCUMENT - FD-TC-0011